

Кирил Кискинов, ИИ0014, група 9

Упражнение 2: Матковски оценки
Таблица 1

№	X	Y	Z
1	25	64	30
2	28	63	48
3	22	65	35
4	20	62	47

$\bar{x} =$	$\bar{y} = 63,5$	$\bar{z} = 40$	$\bar{x} = 23,75$
$S_x^2 =$	$S_y^2 = 1,67$	$S_z^2 = 79,33$	$S_x^2 = 12,25$
$S_x =$	$S_y = 1,29$	$S_z = 8,91$	$S_x = 3,5$
r_{xy}	$\hat{r}_{xz} = 0,04$	$\hat{r}_{yz} = -0,78$	$\hat{r}_{xy} = 0,11$

a) математическо очакване:

а) математическое ожидание:

$$\bar{x} = \frac{25+28+22+20}{4} = \frac{95}{4} = 23,75$$

$$\bar{y} = \frac{64+63+65+62}{4} = \frac{254}{4} = 63,5 \quad \bar{z} = \frac{30+48+35+47}{4} = \frac{160}{4} = 40$$

б) дисперсия:

$$S_x^2 = \frac{(25-23,75)^2 + (28-23,75)^2 + (22-23,75)^2 + (20-23,75)^2}{4-1} =$$
$$= \frac{1,5625 + 18,0625 + 3,0625 + 14,0625}{3} = \frac{36,75}{3} = 12,25$$

$$S_y^2 = \frac{(64-63,5)^2 + (63-63,5)^2 + (65-63,5)^2 + (62-63,5)^2}{4-1} =$$

$$= \frac{0,25 + 0,25 + 2,25 + 2,25}{3} = \frac{5}{3} = 1,6\bar{7}$$

$$S_z^2 = \frac{(30-40)^2 + (48-40)^2 + (35-40)^2 + (47-40)^2}{4-1} = \frac{100+64+25+49}{3} = \frac{238}{3} = 79,33$$

Средноквадратично отклонение:

$$S_x = \sqrt{S_x^2} = \sqrt{12,25} = 3,5 \quad S_y = \sqrt{1,67} = 1,29 \quad S_z = \sqrt{79,33} = 8,91$$

2) коэффициент на Корреляция:
-характери 1 и 2:

$$\hat{r}_{xy} = \frac{(25-23,75)(64-63,5) + (28-23,75)(63-63,5) + (22-23,75)(65-63,5) + (20-23,75)(62-63,5)}{(4-1) \cdot 3,5 \cdot 1,29}$$

$$\hat{r}_{xy} = \frac{(1,25 \cdot 0,5) + (4,25 \cdot (-0,5)) + (-1,75 \cdot 1,5) + (-3,75 \cdot (-1,5))}{3 \cdot 3,5 \cdot 1,29} =$$

$$= \frac{0,625 + -2,125 - 2,625 + 5,625}{13,545} = \frac{1,5}{13,545} = 0,1107$$

$$r_{xy} = \frac{(1,25 \cdot 0,5) + (-2,125 \cdot 0,5) + (-2,625 \cdot 0,5) + (5,625 \cdot 0,5)}{3 \cdot 3,5 \cdot 1,29} =$$

$$3 \cdot 3,5 \cdot 1,29$$

$$= \frac{0,625 - 2,125 - 2,625 + 5,625}{13,545} = \frac{1,5}{13,545} = 0,1107$$

-характери 1 и 3

$$r_{xz} = \frac{(25-23,75)(30-40) + (28-23,75)(48-40) + (22-23,75)(35-40) + (20-23,75)(47-40)}{(4-1) \cdot 3,5 \cdot 8,91}$$

$$r_{xz} = \frac{(1,25 \cdot -10) + (4,25 \cdot 8) + (-1,75 \cdot -5) + (-3,75 \cdot 7)}{3 \cdot 3,5 \cdot 8,91} =$$

$$= \frac{-12,5 + 34 + 8,75 - 26,25}{93,555} = \frac{4}{93,555} = 0,04275$$

-корреля 2 и 3

$$\hat{r}_{yz} = \frac{(64-63,5)(30-40) + (63-63,5)(48-40) + (65-63,5)(35-40) + (62-63,5)(47-40)}{(4-1) \cdot 1,29 \cdot 8,91}$$

$$= \frac{(0,5 \cdot -10) + (-0,5 \cdot 8) + (1,5 \cdot -5) + (-1,5 \cdot 7)}{3 \cdot 1,29 \cdot 8,91} = \frac{-5 - 4 - 7,5 - 10,5}{34,4817} = \frac{-27}{34,4817}$$

$$\hat{r}_{yz} = -0,7830$$

Пример 2

X	102	99	115	120
Y	68	65	79	75
Z	36	48	32	56

$\bar{x} = 124$	$\bar{y} = 71,75$	$\bar{z} = 43$
$S_x^2 = 1242$	$S_y^2 = 40,92$	$S_z^2 = 121,33$
$S_x = 35,24$	$S_y = 6,40$	$S_z = 11,01$
	$\hat{r}_{yz} = -0,21$	

матрица 2

X	102	99	115	120
Y	68	65	79	75
Z	36	48	32	56

$\bar{x} = 124$	$\bar{y} = 71,75$	$\bar{z} = 43$
$S_x^2 = 1242$	$S_y^2 = 40,92$	$S_z^2 = 121,33$
$S_x = 35,24$	$S_y = 6,40$	$S_z = 11,01$
	$\hat{r}_{yz} = -0,21$	

2) коэффициент

$$\hat{r}_{yz} = \frac{68 - 71,75}{11,01} = -0,34$$

$$\hat{r}_{yz} = -0,34$$

$$= \frac{-45}{211,392}$$

а) математическое ожидание:

$$\bar{x} = \frac{102 + 99 + 115 + 120}{4} = \frac{496}{4} = 124 \quad \bar{y} = \frac{68 + 65 + 79 + 75}{4} = \frac{287}{4} = 71,75$$

$$\bar{z} = \frac{36 + 48 + 32 + 56}{4} = \frac{172}{4} = 43$$

б) дисперсия

$$S_x^2 = \frac{(102 - 124)^2 + (99 - 124)^2 + (115 - 124)^2 + (120 - 124)^2}{4 - 1} =$$

$$= \frac{484 + 625 + 2601 + 16}{3} = \frac{3726}{3} = 1242$$

No
1
2
3
4

$$S_y^2 = \frac{(68-71,75)^2 + (65-71,75)^2 + (79-71,75)^2 + (75-71,75)^2}{4-1} = \frac{14,0625 + 45,5625 + 52,5625 + 10,5625}{3}$$

$$= 122,75 / 3 = 40,9166$$

$$S_z^2 = \frac{(36-43)^2 + (48-43)^2 + (32-43)^2 + (56-43)^2}{4-1} = \frac{49 + 25 + 121 + 169}{3} = \frac{364}{3} = 121,3333$$

в) средноточечное отклонение:

$$S_x = \sqrt{S_x^2} = \sqrt{35,24} \quad S_y = \sqrt{40,92} = 6,3968 \quad S_z = \sqrt{121,33} = 11,0194$$

г) коэффициента на корреляции между Y и Z

$$r_{yz} = \frac{(68-71,75)(36-43) + (65-71,75)(48-43) + (79-71,75)(32-43) + (75-71,75)(56-43)}{(4-1) \cdot 6,40 \cdot 11,01}$$

$$r_{yz} = \frac{(-3,75 \cdot -7) + (-6,75 \cdot 5) + (7,25 \cdot -11) + (3,25 \cdot 13)}{3 \cdot 6,40 \cdot 11,01} = \frac{-26,25 + -33,75 - 79,75 + 42,25}{211,392}$$

$$= \frac{-45}{211,392} = -0,212874$$

Таблица 3

№	X	Y	Z
1	35	18	50
2	38	23	46
3	42	15	55
4	40	20	47

$\bar{X} = 38,75$	$\bar{Y} = 19$	$\bar{Z} = 50$
$S_x^2 = 8,92$	$S_y^2 = 11,33$	$S_z^2 = 12,67$
$S_x = 2,99$	$S_y = 3,37$	$S_z = 3,56$
$\hat{r}_{xy} = -0,36$	$\hat{r}_{yz} = -0,86$	

$$\bar{Z} = \frac{50+48+55+47}{4} = \frac{200}{4} = 50$$

а) математическое ожидание

$$\bar{X} = \frac{35+38+42+40}{4} = \frac{155}{4} = 38,75$$

$$\bar{Y} = \frac{18+23+15+20}{4} = \frac{76}{4} = 19$$

б) дисперсия:

$$S_x^2 = \frac{(35-38,75)^2 + (38-38,75)^2 + (42-38,75)^2 + (40-38,75)^2}{4-1} = \frac{14,0625 + 0,5625 + 10,5625 + 1,5625}{3} = \frac{26,75}{3} = 8,91666$$

$$S_y^2 = \frac{(18-19)^2 + (23-19)^2 + (15-19)^2 + (20-19)^2}{4-1} = \frac{1 + 16 + 16 + 1}{3} = \frac{34}{3} = 11,3333$$

$$S_z^2 = \frac{(50-50)^2 + (48-50)^2 + (55-50)^2 + (47-50)^2}{4-1} = \frac{0 + 4 + 25 + 9}{3} = \frac{38}{3} = 12,6666$$

в) средноквадратическое отклонение:

$$S_x = \sqrt{S_x^2} = \sqrt{8,92} = 2,9866$$

$$S_y = \sqrt{11,33} = 3,3660$$

$$S_z = \sqrt{12,67} = 3,5594$$

4-1 3 3
Корреляционно отклонение:

$$S_x = \sqrt{S_x^2} = \sqrt{8,92} = 2,9866 \quad S_y = \sqrt{11,33} = 3,3660 \quad S_z = \sqrt{12,67} = 3,5594$$

2) коэффициент на Корелация:

- Мина 1 и Мина 2:

$$\begin{aligned} \hat{r}_{xy} &= \frac{(35-38,75)(18-19) + (38-38,75)(23-19) + (42-38,75)(15-19) + (40-38,75)(20-19)}{(4-1) \cdot 2,99 \cdot 3,37} \\ &= \frac{-11}{30,2289} = -0,36389 \end{aligned}$$

- Мина 2 и Мина 3

$$\begin{aligned} \hat{r}_{yz} &= \frac{(18-19)(50-50) + (23-19)(48-50) + (15-19)(55-50) + (20-19)(47-50)}{(4-1) \cdot 3,37 \cdot 3,56} \\ &= \frac{-31}{35,9916} = -0,861312 \end{aligned}$$

Таблица 4

X	55	68	62	59
Y	43	45	38	36
Z	80	90	75	68

$\bar{X} = 61$ $\bar{Y} = 40,5$ $\bar{Z} = 78,25$ а) математическое ожидание:

$$S_x^2 = 30 \quad S_y^2 = 17,67 \quad S_z^2 = 85,58$$

$$S_x = 5,48 \quad S_y = 4,20 \quad S_z = 9,25$$

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{55+68+62+59}{4} = \frac{244}{4} = 61$$

$$\bar{Y} = \frac{43+45+38+36}{4} = \frac{162}{4} = 40,5$$

$$\bar{Z} = \frac{80+90+75+68}{4} = \frac{313}{4} = 78,25$$

б) дисперсия: $S_x^2 = \frac{(55-61)^2 + (68-61)^2 + (62-61)^2 + (59-61)^2}{4-1} = \frac{90}{3} = 30$

$$S_y^2 = \frac{(43-40,5)^2 + (45-40,5)^2 + (38-40,5)^2 + (36-40,5)^2}{4-1} = \frac{53}{3} = 17,666 \approx 17,67$$

$$S_z^2 = \frac{(80-78,25)^2 + (90-78,25)^2 + (75-78,25)^2 + (68-78,25)^2}{4-1} = \frac{256,75}{3} = 85,58333 \approx 85,58$$

в) средние квадратичные отклонения: $S_x = \sqrt{S_x^2} = \sqrt{30} = 5,4772 \approx 5,48$

$$S_y = \sqrt{17,67} = 4,2035 \approx 4,20 \quad S_z = \sqrt{85,58} = 9,2509 \approx 9,25$$

г) коэффициенты на корреляцию между X и Z

$$r_{xz} = \frac{(55-61)(90-78,25) + (68-61)(90-78,25) + (62-61)(75-78,25) + (59-61)(68-78,25)}{(4-1) \cdot 5,48 \cdot 9,25}$$

$$r_{xz} = \frac{99}{152,07} = 0,585256 \approx 0,59$$